Development and Verification of a 5G-AKA Security Inspection Tool Based on UERANSIM*

Eensuh Jung, Bumzong Sim, Seongmin Na, Jieon Oh, Bonam Kim and Ilsun You
Kookmin University, Seoul, Republic of Korea
{briel0204,sbz5842,sktjdals0327,jieon1006,
kimbona,isyou}@kookmin.ac.kr

Abstract

This paper proposes a security inspection tool for verifying the 5G-AKA authentication procedure in an Open5GS environment. The proposed tool was implemented by modifying the UE module of UERANSIM to transmit abnormal Authentication Response messages containing manipulated RES* values. Experimental results show that Open5GS successfully detected the abnormal authentication response and rejected the registration procedure. These results confirm that Open5GS properly validates authentication messages and maintains the security policy of the 5G-AKA procedure.

* Proceedings of The 2026 IFIP WG 8.4 International Symposium on E-Business Information Systems Evolution (EBISION 2026), Article No. P-14, July 8-10, 2026, Seoul, South Korea.
© The copyright of this paper remains with the author(s).

1 Introduction

5G is a next-generation mobile communication technology based on ultra-high speed, ultra-low latency, and massive connectivity, and is widely used in industrial fields such as autonomous driving, smart factories, and IoT [1]. Since the 5G Core is based on Control Plane architecture composed of multiple NFs (Network Functions), repetitive authentication requests or abnormal messages may cause security issues such as resource exhaustion and DoS (Denial of Service) [2].

In this paper, we propose a security inspection tool for verifying the 5G-AKA authentication procedure [3] in an Open5GS and UERANSIM environment. The tool was implemented by modifying the NAS authentication messages of the UERANSIM UE to generate abnormal Authentication Response messages. Through this approach, we verified the handling of abnormal authentication messages and the security behavior of the 5G Core.

2 Proposed Security Inspection Method and Implementation

2.1 Inspection Objectives and Test Scenario

This experiment verified whether Open5GS detects and rejects a manipulated Authentication Response when the UE intentionally modifies the RES* value during 5G-AKA. Since improper RES* validation may lead to authentication bypass or illegitimate UE registration, this scenario is important for evaluating basic authentication security.

The test procedure executed the Open5GS 5GC, UERANSIM gNB, and modified UERANSIM UE. The UE transmitted an abnormal Authentication Response after RES* manipulation, and the AMF response was analyzed. Returning Registration Reject was defined as normal behavior, while accepting the UE registration was defined as abnormal behavior.

2.2 Architecture of the Proposed Tool

To generate abnormal authentication messages, the UERANSIM UE capable of simulating real UE behavior was utilized. The modified UERANSIM UE performs the normal Registration procedure but intentionally manipulates the last bit of the RES* value during the Authentication Response generation stage defined in the NAS protocol, thereby generating an invalid RES*. The manipulated Authentication Response message is then transmitted to the Open5GS AMF for security verification.

2.3 Development and Verification Results of the Inspection Tool

The verification environment used Open5GS to simulate AMF, SMF, and UPF functions, while UERANSIM simulated the gNB and UE. This configuration reproduced the 5G Registration and Authentication procedure.

Experimental results showed that the Open5GS AMF detected the mismatch between the manipulated RES* and the valid authentication value, then returned Registration Reject to block UE registration. Therefore, Open5GS properly validates RES* during 5G-AKA and is considered secure against this authentication response tampering scenario.

3 Conclusion

In this paper, we proposed a UERANSIM-based security inspection tool for verifying authentication response tampering in an Open5GS-based 5G Core environment. The proposed tool was implemented by modifying the RES* value generated by the UERANSIM UE to transmit abnormal Authentication Response messages during the 5G-AKA procedure.

Experimental results showed that the Open5GS AMF detected the manipulated RES* value and returned a Registration Reject message, thereby blocking UE registration. This confirms that Open5GS properly validates and rejects manipulated authentication responses during the 5G-AKA procedure.

The proposed tool can be utilized in 5G private network environments to verify authentication bypass attempts, abnormal UE access control, and security configuration validation. Since this study focused on a single RES* last-bit tampering case, future work will diversify the experimental scenarios by applying various RES* tampering patterns and extending the tool to additional scenarios such as GUTI reallocation, NAS message tampering, and Registration request-based DoS testing.

Acknowledgement: This work was supported by Institute of Information & communications Technology Planning & Evaluation (IITP) grant funded by the Korea government (MSIT) (RS-2024-00441484, Development of open roaming technology for Private 5G network).

References

[1] 3GPP TS 22.261, “Service requirements for the 5G system”, V15.9.0.
[2] KISA, (2026), “Security Guidelines for 5G Private Networks”.
[3] 3GPP TS 33.501, “Security architecture and procedures for 5G system”, V15.9.0.